

## SAN PEDRO, Côte d'Ivoire

WMO: 655940

Lat: 4.747N

Lon: 6.661W

Elev: 26

StdP: 14.68

Time Zone: 0.00 (E00)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |       |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|-------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |       |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR    | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)   | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 70.7       | 72.0 | 63.7                          | 88.3 | 76.3 | 68.0 | 103.1 | 76.1 | 21.2                  | 79.4 | 18.8 | 78.7 | 3.0                   | 0    | 0.326 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 11.6                   | 89.9            | 80.2 | 89.3 | 80.3 | 88.3 | 80.1 | 82.6                | 87.1 | 81.9 | 86.6 | 81.4 | 86.2 | 8.8                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.0                            | 161.2 | 85.3 | 80.7 | 159.5 | 85.1 | 80.1 | 156.5 | 84.7 | 46.4          | 87.9 | 45.7 | 87.0 | 45.1 | 86.5 | 85.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 17.7          | 14.8 | 13.4 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 80.2   | 81.1  | 81.9  | 82.5  | 82.5  | 80.9  | 79.2  | 77.9  | 77.0  | 78.2  | 79.7  | 80.4  | 80.8  |
|                                                                                  | DBStd       | 2.31   | 1.71  | 1.76  | 1.67  | 1.72  | 1.82  | 1.64  | 1.31  | 1.40  | 1.31  | 1.44  | 1.26  | 1.25  |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD50       | 11007  | 964   | 892   | 1007  | 974   | 958   | 876   | 864   | 837   | 847   | 922   | 912   | 954   |
|                                                                                  | CDD65       | 5532   | 499   | 472   | 542   | 524   | 493   | 426   | 399   | 372   | 397   | 457   | 462   | 489   |
|                                                                                  | CDH74       | 51545  | 5065  | 4963  | 5962  | 5989  | 5106  | 3615  | 2668  | 1978  | 2590  | 4006  | 4615  | 4988  |
|                                                                                  | CDH80       | 13989  | 1660  | 1707  | 2168  | 2204  | 1514  | 572   | 227   | 118   | 277   | 730   | 1261  | 1551  |
| Wind                                                                             | WSAvg       | 6.0    | 5.1   | 5.7   | 6.3   | 6.8   | 5.8   | 6.6   | 6.9   | 7.1   | 6.6   | 6.4   | 4.9   | 4.3   |
| Precipitation                                                                    | PrecAvg     | 51.9   | 0.8   | 1.3   | 2.2   | 4.2   | 10.4  | 13.0  | 4.1   | 2.6   | 2.6   | 5.0   | 5.2   | 2.2   |
|                                                                                  | PrecMax     | 81.5   | 2.3   | 5.0   | 7.8   | 12.8  | 27.9  | 21.5  | 17.4  | 7.6   | 8.7   | 15.0  | 10.8  | 6.8   |
|                                                                                  | PrecMin     | 31.9   | 0.0   | 0.0   | 0.0   | 0.3   | 3.7   | 3.8   | 0.2   | 0.2   | 0.0   | 0.3   | 0.6   | 0.0   |
|                                                                                  | PrecStd     | 12.3   | 0.7   | 1.5   | 1.9   | 2.9   | 6.7   | 4.9   | 4.5   | 1.8   | 2.4   | 3.6   | 2.8   | 1.7   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 89.9  | 90.3  | 91.1  | 90.8  | 89.1  | 86.4  | 84.2  | 83.5  | 84.6  | 86.3  | 88.2  | 89.6  |
|                                                                                  |             | MCWB   | 77.5  | 79.9  | 80.7  | 81.2  | 80.1  | 79.5  | 76.6  | 76.0  | 77.5  | 78.6  | 79.9  | 78.8  |
|                                                                                  | 2%          | DB     | 88.6  | 89.3  | 89.9  | 89.8  | 88.0  | 85.4  | 82.9  | 82.4  | 83.6  | 85.4  | 87.1  | 88.0  |
|                                                                                  |             | MCWB   | 78.6  | 80.1  | 80.7  | 81.2  | 80.3  | 79.0  | 76.1  | 75.6  | 77.2  | 78.4  | 79.7  | 79.2  |
|                                                                                  | 5%          | DB     | 87.8  | 88.3  | 89.0  | 88.9  | 87.1  | 84.2  | 82.3  | 81.3  | 82.6  | 84.5  | 86.2  | 87.1  |
|                                                                                  |             | MCWB   | 78.7  | 80.0  | 80.6  | 81.0  | 80.1  | 78.3  | 75.9  | 75.3  | 76.9  | 78.0  | 79.3  | 79.1  |
|                                                                                  | 10%         | DB     | 86.8  | 87.4  | 87.9  | 87.9  | 86.0  | 83.0  | 81.4  | 80.5  | 82.0  | 83.9  | 85.5  | 86.2  |
|                                                                                  |             | MCWB   | 78.7  | 79.8  | 80.4  | 80.7  | 79.7  | 77.7  | 75.6  | 75.1  | 76.7  | 77.8  | 79.0  | 79.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 81.9  | 82.7  | 82.9  | 83.5  | 82.8  | 81.3  | 78.5  | 78.6  | 79.2  | 80.0  | 81.6  | 81.9  |
|                                                                                  |             | MCDB   | 86.5  | 87.4  | 89.0  | 87.9  | 86.2  | 84.3  | 81.1  | 80.4  | 81.9  | 83.8  | 86.2  | 86.2  |
|                                                                                  | 2%          | WB     | 80.9  | 81.8  | 82.3  | 82.6  | 81.6  | 79.9  | 77.6  | 77.4  | 78.3  | 79.2  | 80.5  | 80.8  |
|                                                                                  |             | MCDB   | 86.0  | 86.6  | 87.6  | 87.1  | 85.4  | 83.2  | 80.6  | 79.7  | 81.5  | 83.4  | 85.5  | 86.0  |
|                                                                                  | 5%          | WB     | 80.3  | 81.0  | 81.6  | 81.8  | 80.9  | 79.1  | 77.0  | 76.5  | 77.6  | 78.8  | 79.9  | 80.1  |
|                                                                                  |             | MCDB   | 85.4  | 86.0  | 86.8  | 86.6  | 85.1  | 82.7  | 80.4  | 79.2  | 80.9  | 83.2  | 84.8  | 85.3  |
|                                                                                  | 10%         | WB     | 79.8  | 80.5  | 81.0  | 81.3  | 80.2  | 78.5  | 76.4  | 75.7  | 77.1  | 78.3  | 79.2  | 79.6  |
|                                                                                  |             | MCDB   | 84.8  | 85.6  | 86.0  | 86.2  | 84.6  | 82.0  | 79.8  | 78.7  | 80.6  | 82.4  | 84.2  | 84.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 13.1  | 12.5  | 12.3  | 11.6  | 10.1  | 7.9   | 7.5   | 7.4   | 7.7   | 8.9   | 10.8  | 12.1  |
|                                                                                  |             | MCDBR  | 13.7  | 13.1  | 13.2  | 12.6  | 11.5  | 9.3   | 8.4   | 8.6   | 8.7   | 9.6   | 11.8  | 13.2  |
|                                                                                  |             | MCWBR  | 6.2   | 5.6   | 5.8   | 5.6   | 5.2   | 4.4   | 3.5   | 3.9   | 3.9   | 4.4   | 5.6   | 5.9   |
|                                                                                  | 5% WB       | MCDBR  | 12.3  | 12.1  | 12.3  | 12.0  | 10.6  | 8.8   | 7.6   | 7.7   | 8.2   | 9.2   | 11.5  | 12.2  |
|                                                                                  | MCWBR       | 5.9    | 5.7   | 5.9   | 6.0   | 5.6   | 4.9   | 3.5   | 4.2   | 4.2   | 4.5   | 5.9   | 6.0   |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.755 | 0.716 | 0.717 | 0.557 | 0.496 | 0.506 | 0.510 | 0.498 | 0.470 | 0.464 | 0.516 | 0.644 |
|                                                                                  | taud        |        | 1.604 | 1.694 | 1.693 | 2.059 | 2.234 | 2.190 | 2.159 | 2.201 | 2.313 | 2.348 | 2.157 | 1.814 |
|                                                                                  | Ebn at Noon |        | 199   | 212   | 213   | 246   | 255   | 248   | 249   | 257   | 269   | 272   | 256   | 221   |
|                                                                                  | Edh at Noon |        | 86    | 80    | 80    | 55    | 44    | 46    | 48    | 47    | 43    | 41    | 49    | 69    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1571  | 1542  | 1655  | 1643  | 1431  | 1245  | 1352  | 1364  | 1454  | 1577  | 1565  | 1503  |
|                                                                                  | RadStd      |        | 81    | 73    | 84    | 83    | 80    | 95    | 105   | 95    | 63    | 63    | 57    | 49    |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page